

Module 24.12: ElasticNet Overview (Hybrid L1 + L2 Regularization)

1. Limitations of Pure Lasso and Pure Ridge

While Lasso and Ridge are foundational, both suffer from distinct operational limitations in high-dimensional real-world settings:

Scenario	Lasso Limitation	Ridge Limitation
Collinear Feature Groups	Arbitrarily selects one feature from a correlated cluster and drops the rest to zero, discarding group signal.	Retains all features with shared small weights; cannot perform feature selection.
High Dimensions ($p > m$)	Can select at most m features before saturating, regardless of how many true predictive signals exist.	Retains all p features, risking overfitting.

2. Mathematical Formulation of ElasticNet

Introduced by Zou and Hastie (2005), **ElasticNet** synthesizes both penalties into a single convex objective:

$$J_{\text{ElasticNet}}(\mathbf{w}, b) = \frac{1}{2m} \sum_{i=1}^m \left(\hat{y}^{(i)} - y^{(i)} \right)^2 + \alpha \left[\rho |\mathbf{w}|_1 + \frac{1-\rho}{2} |\mathbf{w}|_2^2 \right]$$

Where:

- $\alpha \geq 0$: Total regularization strength.
- $\rho \in [0, 1]$ (the `l1_ratio` parameter): Balances the mixture between L1 and L2 penalties:
 - $\rho = 1.0 \implies$ Pure Lasso Regression.
 - $\rho = 0.0 \implies$ Pure Ridge Regression.

- $0 < \rho < 1.0 \implies$ Hybrid ElasticNet.

3. The "Grouping Effect" Advantage

ElasticNet provides the **grouping effect**:

- Strongly correlated features share similar regression coefficients and enter or leave the model together as a group (inherited from L2).
- Simultaneously, entire groups of irrelevant features are cleanly set to zero (inherited from L1).
- The geometry of ElasticNet's constraint contour combines the sharp vertices of the L1 diamond with the strictly convex curvature of the L2 sphere, eliminating non-unique solutions.